

EMILY DEEB

Spatial Data Scientist | Urban Analytics, Design & Consultancy | MSc Urban Spatial Science, CASA UCL
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PROFESSIONAL SUMMARY

Geospatial data scientist and licensed architect who turns spatial data into design decisions and client-ready recommendations. Brings five years of urban design and GIS leadership in practice, delivering evidence-based plans for municipal and private clients, from city-centre regeneration to a 100,000-unit city expansion. Holds an MSc in Urban Spatial Data Science from CASA, UCL, including an industry dissertation with CACI Ltd that built Python machine learning pipelines on national consumer spending data. Combines statistical modelling, GIS and machine learning with design thinking and stakeholder communication to take analysis from model output to real-world outcome.

PROFESSIONAL EXPERIENCE

CACI Ltd (GeoDS Industry Programme), London | May 2026 – Sep 2026

Spatial Data Scientist — Industry Dissertation

- Investigated whether urban morphological fabric type predicts retail spending patterns across UK cities, independently of demographic composition, using CACI proprietary consumer spending data and HiMoC morphometric classifications.
- Built an end-to-end Python machine learning pipeline using GeoPandas, scikit-learn, SHAP and spatial autocorrelation analysis, applying pre-modelling diagnostics, feature engineering and model evaluation to produce reproducible, documented outputs.
- Found that physical form explained 30.1% of predictive variance beyond demographics across all six retail spending categories, with street-network betweenness centrality the strongest single physical predictor ($r = 0.33$).
- Framed findings for commercial application, showing how urban form and street-network metrics could strengthen location models used in retail site selection and catchment analysis.
- Presented findings to CACI's Consumer and Location Intelligence team and at the GeoDS Data Partner Forum, Saïd Business School, Oxford.

Buro Happold, London | Jun 2026

Industry Consultant — Building a Better Future Scheme

- Led research scoping and coordinated a multidisciplinary team of six, managing stakeholder expectations and synthesising meeting outputs into a clear project direction.
- Delivered a final report, client presentation and a structured cloud repository of research and data to support Buro Happold's future work.

Derman Verbakel Architecture, Tel Aviv | Sep 2020 – Sep 2025

Team Leader — Urban Design & GIS

- Translated geospatial analysis into design and planning decisions for municipal and private clients, leading a team of 4 across 15 concurrent projects from block to city scale.
- Built the evidence base for Sakhnin city centre regeneration (1,200 units) from scratch by collecting and integrating multi-source spatial datasets. Turned the findings into design proposals, spatial reports and cartographic visualisations for municipal planning committees.
- Developed QGIS and ArcGIS workflows for site feasibility, market analysis and profitability modelling that informed site and density decisions for clients, improving office efficiency by 30%.

- Trained 15+ team members in GIS tools; developed reproducible, documented spatial workflows adopted across the office.
- Key projects: Transportation Masterplan of Eilat; Regeneration of Sakhnin City Centre (1,200 units); TOD Regeneration Plan Rishon LeTsiyon (3,000 units); City expansion masterplan (100,000 units, confidential).

SELECTED PROJECTS

London Crime Density & Urban Amenities Analysis | 2025

Python, Machine Learning, Spatial Analysis, London Datastore

- Analysed 1.78 million Metropolitan Police Service crime records across 679 London wards to investigate relationships between urban amenity distribution and crime density.
- Applied multiple regression, K-Nearest Neighbours (KNN), and Random Forest models with SHAP decomposition; found nightlife amenities strongest predictors of person/society crime, retail nodes driving property crime.

Food Shock — Global Food Supply Crisis Tracker | 2026

Group Project — React, Leaflet, FAO Data, Python

- Built an interactive web application visualising global food supply chain disruptions using FAO trade flow data (200 countries, 82 commodities), commodity price indices, and spatial mapping.

EDUCATION

MSc Urban Spatial Science | Sep 2025 – Oct 2026

The Centre for Advanced Spatial Analysis (CASA), UCL

- Specialised in computational urban analysis, spatial data science, geocomputation, statistical modelling, spatial statistics, urban simulation, data visualisation, and machine learning in Python and R.

B.Arch. Architecture | Sep 2015 – Jun 2020

Bezalel Academy of Arts and Design Jerusalem

- Graduation Project: Destination: Jericho — research-led urban design study using QGIS, spatial analysis, and data visualisation. Nominated for David Azrieli Architecture Prize.

TECHNICAL SKILLS

Programming & Data: Python (Pandas, NumPy, Matplotlib, GeoPandas, Scikit-learn), R (RStudio), SQL, PostGIS, React, Leaflet.

Data Science & ML: Machine Learning (ML), Classification, Clustering, Regression, Random Forest, KNN, Feature Engineering, Model Evaluation, Statistical Analysis, Spatial Statistics, SHAP, Spatial Autocorrelation, Pre-modelling Diagnostics, Predictive Modelling.

Data & Viz: Data Pipelines, Data Cleaning, Reproducible Workflows, Data Visualisation, Dashboards, Power BI, Quarto, Jupyter, Git.

Spatial & GIS: GIS (Geographic Information Systems), Geospatial Analysis, QGIS, ArcGIS, Spatial Network Analysis, Spatial Data Science, Urban Morphology, Cartography, Google Earth Engine API, Remote Sensing.

Design & Other: AutoCAD, Rhinoceros, Adobe Suite (Photoshop, Illustrator, InDesign), Microsoft Office (Excel, Word, PowerPoint), UI/UX (CalArts certified).

Consultancy & Practice: Evidence-based Design, Urban Planning, Stakeholder Engagement, Client Reporting, Feasibility Studies, Masterplanning, Team Leadership.

LANGUAGES & CERTIFICATIONS

English (fluent), Hebrew (fluent), Arabic (native) | Licensed Architect — Israeli Labor Department, 2025.

UK right to work: Graduate visa (valid until 2029), no sponsorship required.